

Harder negative sets lower apparent AUROC and raise measured model contribution: a three-protocol calibration across 94 ENCODE eCLIP datasets

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Abstract

Sequence models for RNA-binding proteins are evaluated against negative windows, and how those windows are chosen is usually treated as an implementation detail. Across 94 paired ENCODE eCLIP datasets we hold the model, the positives, the chromosome-blocked folds and the estimator fixed and vary only the negative-set protocol, measuring the nested contribution of a 4-mer model over a 19-feature mono- and dinucleotide composition baseline refit under each protocol. Replacing GC-matched with dinucleotide-matched negatives lowers apparent AUROC by 0.110 in 94 of 94 datasets while raising the measured contribution from 0.027 to 0.066. A bias-aware protocol drawing negatives from other proteins’ binding sites is the easiest of the three discriminations and yields the least, 0.012. Apparent difficulty and measured contribution therefore move in opposite directions, spanning 5.4-fold for a 4-mer model and between 3.8 and 7.6-fold across three model classes measured on identical rows, a range about three times the gap between a k-mer model and a convolutional network on the same datasets. The composition baseline each protocol leaves behind accounts for most of the variation for the 4-mer, though not for every model class. No rescaling supplies a transportable remedy: an exponent that equalises our three protocols exists, but the exponent fitted to an independent benchmark differs from it 2.4-fold. We recommend reporting the composition-only AUROC obtained under the same protocol beside every headline AUROC, and not comparing contributions measured under different protocols.

Keywords: RNA-binding proteins; eCLIP; benchmarking; negative-set construction; sequence composition; incremental value; AUROC

1 Introduction

Sequence models that predict RNA-binding protein (RBP) occupancy are evaluated almost entirely by the area under the receiver operating characteristic curve (AUROC) on held-out windows. Positive windows come from a crosslinking assay, most often eCLIP [2, 3], but negative windows have to be constructed, and the construction is usually given a sentence in the methods. Horlacher

et al. [4] benchmarked eleven RBP prediction methods across hundreds of CLIP-seq experiments under two negative-set definitions and showed that a method’s apparent performance depends on which definition is used. They proposed a bias-aware alternative in which negatives are drawn from the binding sites of other RBPs rather than from genomic background.

How large that dependence is, and in which direction it acts, remains open. A raw AUROC conflates the signal a model found with the difficulty of the discrimination it was set, and whoever builds the benchmark controls the second. Negatives sampled uniformly from the genome differ from crosslinked positives in base composition, so a classifier responding to nucleotide frequency alone will score well without having learned anything sequence-specific. Matching negatives on GC content is the usual mitigation, but GC content is one linear functional of the sixteen-cell dinucleotide simplex, leaving the remaining fourteen degrees of freedom of compositional difference available to any model. Comparable circularity is characterised in variant-effect prediction, where the assembly of the evaluation set both inflates apparent accuracy and reorders methods [6], and the sensitivity of genomic machine-learning evaluations to the definition of the negative class is recognised more generally [7].

A model’s increment over an explicit composition baseline is less exposed to compositional confounding than its raw AUROC. We define the nested contribution as the difference between the pooled out-of-fold AUROC of a logistic model on composition features together with the model’s score, and that of the composition features alone, fitted on the same rows and the same folds.

We assembled 94 ENCODE eCLIP datasets and constructed negatives three ways: matched on GC content, matched on all sixteen dinucleotide frequencies, and drawn from the binding sites of other RBPs assayed in the same cell line, following Horlacher *et al.* [4]. Model, positives, chromosome-blocked folds and estimator were held fixed across arms, and the composition baseline was refit on each arm’s own windows so that no baseline was carried between designs. Three model classes were measured under all three protocols: a 4-mer logistic regression, a 7089-parameter convolutional network, and a fully fine-tuned 19.78 M-parameter nucleotide language model.

Apparent difficulty and measured contribution move in opposite directions. Dinucleotide matching reduced apparent AUROC in all 94 datasets while approximately tripling the nested contribution, and the bias-aware protocol gave both the easiest discrimination of the three and the smallest contribution. The contribution spanned 5.4-fold across protocols for the 4-mer, and between 3.8 and 7.6-fold across the three model classes; by comparison, 0.019 AUROC separates the 4-mer from the convolutional network on this panel. Eight standard monotone reparameterisations reduced the span without closing it, and an exponent fitted to close it did not transport to an independent benchmark. On that basis we recommend reporting the composition-only AUROC alongside every headline AUROC, and we set out what that does and does not achieve.

2 Results

2.1 Effect of negative-set protocol on measured contribution

Across 94 paired datasets the same 4-mer model gave three different accounts of its own contribution depending only on how the negative windows were built (Table 1, Figure 1).

Table 1: Composition-only AUROC, apparent AUROC and nested contribution of a 4-mer model under three negative-set protocols. Model, positives, folds and estimator are identical across arms; $n = 94$ datasets.

protocol	composition alone	apparent AUROC	nested contribution
dinucleotide-matched	0.6274	0.6937	+0.0663
GC-matched	0.7827	0.8092	+0.0265
bias-aware	0.8248	0.8370	+0.0122

Replacing GC-matched with dinucleotide-matched negatives reduced apparent AUROC by 0.1095, in all 94 datasets (paired Wilcoxon $p < 10^{-15}$), while raising the nested contribution from +0.0265 to +0.0663. The contrast in contribution was +0.0398 (95% CI +0.0325 to +0.0477, protein-clustered; positive in 88 of 94 datasets). Evaluated by apparent AUROC the model appears to have deteriorated; evaluated by contribution it appears to have improved almost threefold.

The bias-aware protocol did not behave as we predicted. Because both classes are genuine crosslink sites, we expected composition alone to fall toward 0.5 and the contribution to rise. Neither occurred: composition alone was the highest of the three arms at 0.8248, apparent AUROC was the highest at 0.8370, and the contribution was the lowest at +0.0122, or 0.53 times the GC arm (geometric mean over the 83 datasets positive in both arms; the ratio of panel means is 0.46).

Distinct RBPs occupy compositionally distinct sites, so discriminating one protein’s sites from another’s is itself largely a composition task.

Ranked by composition-only AUROC, by apparent AUROC or by contribution, the three protocols order identically. Harder discriminations do reveal more of what a model adds; what does not follow is that a bias-aware protocol is the harder one. Under GC matching, composition alone exceeded the 4-mer’s own AUROC in 29 of 94 datasets, and the contribution was significantly positive in 80 of 94, falling to 72 of 94 at the design effect measured for these data (Methods).

The direction of the GC-versus-dinucleotide contrast is implied by the design. The composition baseline spans the fifteen degrees of freedom of the dinucleotide simplex; GC matching constrains one linear functional of that space and dinucleotide matching constrains all fifteen, so the arm that constrains more must leave the baseline weaker. Only the magnitude of that contrast is informative. The position of the bias-aware arm is not implied by the same argument. That protocol constrains none of the fifteen degrees of freedom, and its negatives are not composition-matched in any respect (median absolute GC difference 0.1387, against 0.0297 for the GC arm). It nonetheless has the highest composition baseline of the three.

Two artefacts were bounded against matched placebos with twenty placebo seeds each. Strand

misassignment, which arises because a matched negative inherits its positive’s strand label, accounted for -0.0055 (CI -0.0089 to -0.0022) of the contrast, leaving 85.4%. Negatives drawn from untranscribed loci accounted for -0.0043 (CI -0.0069 to -0.0018), leaving 89.7%; 40.1% of negatives lie in untranscribed loci, but this fraction is balanced between arms to within 1.2×10^{-5} ($p = 0.64$), and a confound present equally in both arms cannot produce a difference between them. GC-matched negatives are 42.8% sense-strand against 47.4% in the dinucleotide arm, higher in 37 of 40 datasets ($p = 5.0 \times 10^{-9}$), so the spurious directional cue is stronger in the arm with the smaller contribution and acts against the reported direction.

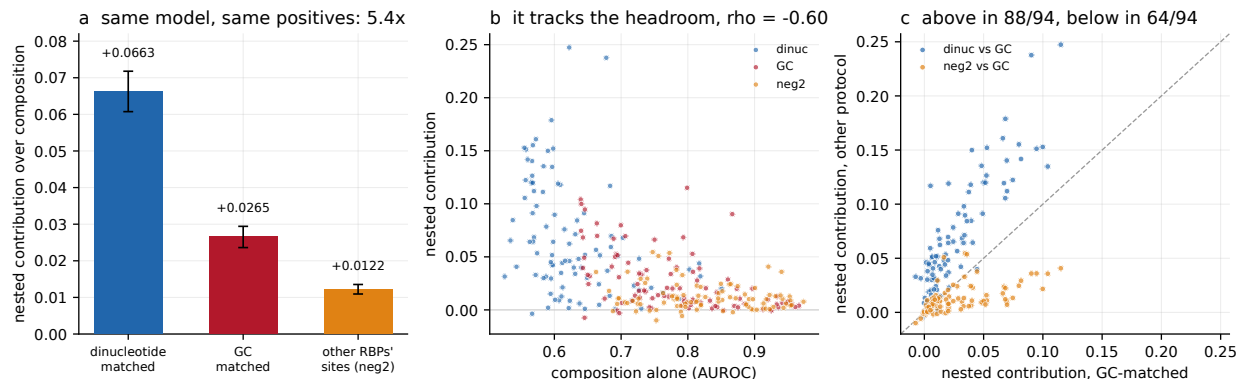


Figure 1: Nested contribution of a 4-mer model under three negative-set protocols, $n = 94$ datasets. (a) Composition-only AUROC, apparent AUROC and nested contribution by protocol. (b) Nested contribution per dataset, dinucleotide-matched against GC-matched; points above the diagonal are the 88 of 94 datasets in which the contribution is greater under dinucleotide matching. (c) The same comparison for the bias-aware arm against GC matching, lower in 64 of 94. Apparent AUROC falls in all 94 datasets under dinucleotide matching while the contribution rises.

2.2 Behaviour under monotone reparameterisation

AUROC is compressive near unity, so a contribution measured against a baseline of 0.82 is arithmetically smaller than the same contribution measured against 0.63, and part of the 5.42-fold span is a property of the scale rather than of the protocol (Figure 2).

Across eight monotone reparameterisations of the same 282 protocol-dataset cells the span was reduced but not eliminated. Somers’ D [10], an affine map of AUROC, returned 5.42 by construction and serves as an internal control. Unbounded links reduced the span in the order in which they stretch the upper tail: arcsine 3.99, complementary log-log 3.94, binormal d' 3.18, logit 2.72. The smallest span, 2.00 (CI 1.67 to 2.46), was obtained by dividing the contribution by the headroom of its own baseline, 1 minus the composition AUROC. Normalising by the excess over chance instead increased the span to 18.2, because that denominator varies with protocol and approaches zero.

A span of 2.00 should not be compared against unity. The ratio of the largest to the smallest of three sample means is bounded below by one and is upward-biased, so some span is expected under any sampling. Simulating equal true arm means while preserving the observed between-arm dataset

pairing gave a null distribution with median 1.064 and 95th percentile 1.175, so the residual span lies outside what sampling alone produces.

Division by headroom is not a distinct coordinate but the $p = 1$ member of the family $g/(1 - c)^p$, where g is the contribution and c the composition AUROC, and the sweep above terminates at the member we recommend. Over the family the span falls to 1.005 at $p = 1.544$ (Table 2), so a monotone rescaling that equalises these three protocols does exist.

Table 2: Fold span across the three protocols for $g/(1 - c)^p$. An exponent that equalises the three protocols exists in sample.

p	0	0.5	1.0	1.25	1.544	1.75	2.0
span	5.42	3.43	2.00	1.48	1.005	1.34	1.95

That exponent is a property of the benchmark rather than of the quantity. Fitted to the two negative sets of Horlacher et al. [4] the equalising exponent is 3.649, and our exponent leaves their benchmark at 2.340 against the 2.381 it started from. A normalisation strong enough to equalise one benchmark therefore has to be refitted on the next.

Extending the sweep to the other two model classes qualifies the headline figure. The floor over the eight reparameterisations is 2.00 for the 4-mer, 2.91 for the convolutional network and 1.68 for SpliceBERT: no model class reaches protocol independence among fixed coordinates and all three exceed the equal-means null, but 2.00 is the 4-mer’s floor rather than a general one. The fitted exponent also performs worse for the neural models, reaching 1.213 for the convolutional network and 1.124 for SpliceBERT against 1.004 for the 4-mer, and the equalising exponent itself differs by a factor of 1.22 between model classes measured on the same benchmark (1.544, 1.843, 1.507).

2.3 Attribution to the composition baseline

Pooled over all 282 protocol-dataset cells the contribution was inversely related to the composition baseline (Spearman -0.600 , $p = 6 \times 10^{-29}$), which is to say that most of what a protocol does is determine how much room the baseline leaves. Three analyses localise this (Figure 3).

Regressing contribution on a cubic in the composition baseline over all 282 cells and then adding protocol indicators, the protocol label contributed 1.0% of variance given the baseline and the baseline contributed 11.0% given the label.

The bias-aware protocol usually raises the baseline relative to GC matching, but lowers it in 27 of 94 datasets, which permits a within-panel test. Where it raised the baseline its contribution deficit was -0.0212 (CI -0.0269 to -0.0157); where it lowered the baseline the deficit reversed to $+0.0028$ (CI -0.0000 to $+0.0063$). Within datasets, the change in baseline and the change in contribution were related at Spearman -0.664 ($p = 3 \times 10^{-13}$).

Matching on the baseline directly is possible only across datasets, because the dinucleotide baseline is below the GC baseline in all 94 datasets and the two are therefore perfectly rank-confounded within a dataset. Under nearest-baseline matching the $+0.0398$ contrast fell to -0.0087

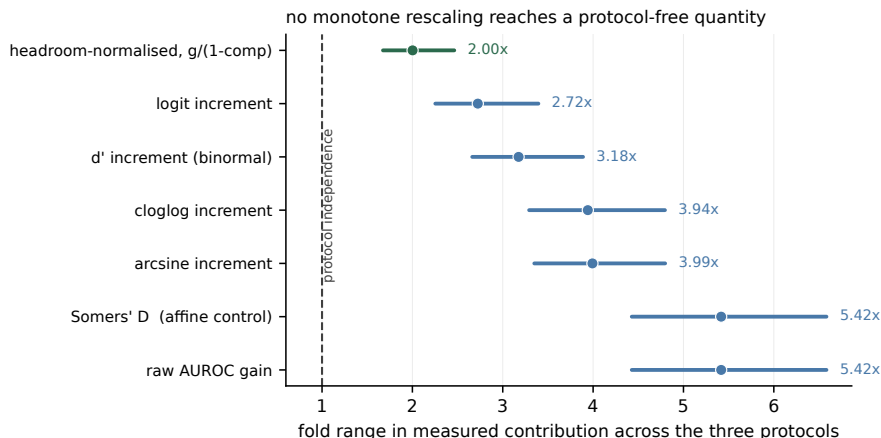


Figure 2: Fold span in nested contribution across the three protocols under eight monotone reparameterisations of the same 282 protocol-dataset cells. Points are the span and bars are 95% percentile intervals from a bootstrap over datasets (2000 draws); the dashed line marks protocol independence. Somers’ D is an affine map of AUROC and returns the raw value by construction, serving as an internal control. The smallest span, 2.00 (1.67–2.46), is obtained by dividing by the headroom of the baseline.

(CI -0.0265 to $+0.0122$, $n = 41$ pairs). This design is not a clean test of the contrast. The retained pairs are drawn from opposing tails of their respective distributions: mean baseline 0.6875 against a panel mean of 0.6274 for the dinucleotide members, and 0.6904 against 0.7827 for their GC partners. The 41 comparisons are served by 24 distinct GC partners, one of them reused eleven times, and proteins are not matched. Nearest-neighbour matching on a covariate monotonically related to the outcome, with replacement and across proteins, is biased toward the null here, and the common support for this pair is 0.073 AUROC wide over 33 of 188 cells.

The GC-versus-bias-aware pair overlaps by 0.212 AUROC over 130 of 188 cells, where the same design is better conditioned. There the bias-aware arm retained a deficit of -0.0081 (CI -0.0130 to -0.0036) under nearest-baseline matching and -0.0043 (CI -0.0077 to -0.0012) as a within-dataset intercept at zero baseline shift. Decomposed by pair, the protocol label contributed 0.05% of incremental variance for GC versus dinucleotide and 3.40% for GC versus bias-aware, so the pooled 1.0% averages a comparison that is not identified with one that is.

Within arms, the relation between baseline and contribution was present for composition-matched negatives and absent otherwise: Spearman -0.545 (GC), -0.462 (dinucleotide) and -0.122 (bias-aware, not significant). This suggests two protocol families rather than three interchangeable protocols, with a residual of 0.005 to 0.008 AUROC between families.

These analyses do not generalise across model classes (Table 3). For the convolutional network the ordering reverses, the protocol label contributing 6.25% of variance against the baseline’s 2.34%. For SpliceBERT the within-arm relation is strong in all three arms, including the bias-aware arm at -0.568 where the 4-mer shows -0.122 , so no two-family structure is apparent in that measurement. The variance attribution, the within-panel reversal and the family structure are properties of the

4-mer measurement and should not be read as properties of the quantity.

Table 3: Attribution of variance in the nested contribution, and the within-arm relation between baseline and contribution, by model class ($n = 94$).

model	incremental R^2		within-arm Spearman		
	protocol	base	base	protocol	dinuc GC bias-aware
4-mer	1.00%		11.05%		−0.462 −0.545 −0.122
CNN	6.25%		2.34%		−0.120 −0.410 +0.075
SpliceBERT	1.07%		24.06%		−0.583 −0.767 −0.568

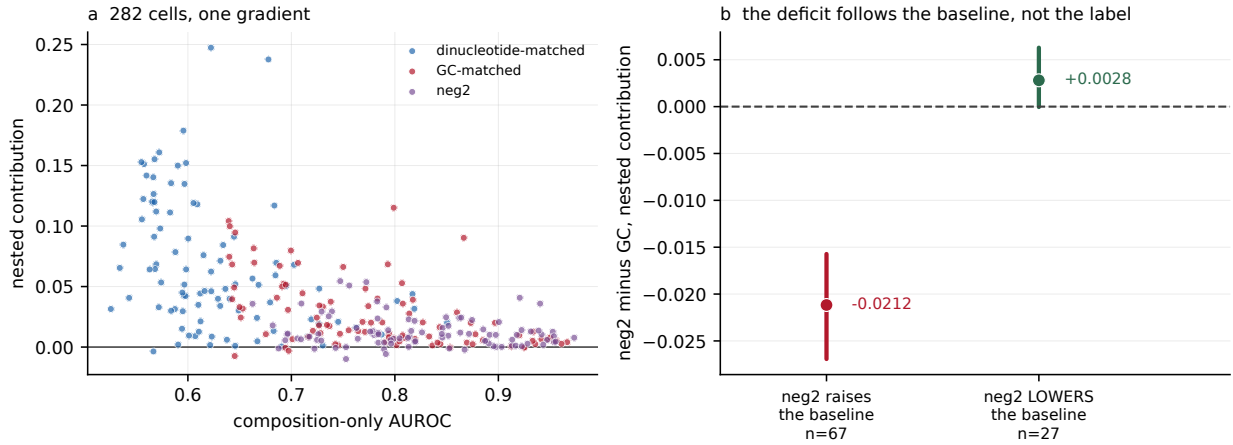


Figure 3: (a) All 282 protocol-dataset cells: nested contribution against the composition-only AUROC left by the protocol, by protocol (pooled Spearman -0.600 , $p = 6 \times 10^{-29}$). (b) Datasets partitioned by whether the bias-aware protocol raised or lowered the composition baseline relative to GC matching ($n = 67$ and 27); points are stratum means and bars are 95% percentile intervals from a bootstrap over datasets (4000 draws). The contribution deficit reverses where the baseline falls.

2.4 Dependence on model class

Both neural models were trained on all three arms under the same folds, seed and code path, with per-window out-of-fold scores retained (Figure 4, Table 4).

Table 4: Nested contribution under three negative-set protocols for three model classes, $n = 94$ datasets, identical rows and folds. Spans are ratios of panel means with protein-clustered percentile intervals.

model	dinucleotide	GC	bias-aware	span
4-mer	+0.0663	+0.0265	+0.0122	5.42 (4.43–6.58)
CNN	+0.0860	+0.0330	+0.0113	7.63 (6.23–9.63)
SpliceBERT	+0.1754	+0.0890	+0.0466	3.76 (3.27–4.34)
composition alone	0.6274	0.7827	0.8248	

The span was present for every model class and was smallest, at 3.76-fold, for the largest model. Across all three, the bias-aware protocol yielded the lowest contribution for all three model classes while carrying the highest composition baseline, so the inversion between apparent difficulty and measured contribution is not specific to a bag-of- k -mers measurement. The two-arm contrast was +0.0398 (CI +0.0325 to +0.0477) for the 4-mer, +0.0530 (CI +0.0440 to +0.0627) for the convolutional network and +0.0864 (CI +0.0778 to +0.0951) for SpliceBERT. On the ratio scale the ordering reverses, with multipliers of 3.08, 3.51 and 2.38 (geometric mean over the 77 datasets positive in both arms for all three models), so the contrast does not increase with model capacity.

Decomposing the log multiplier over the 262 dataset-by-model cells in which both arms were positive, RBP identity accounted for 64.8% of variance, cell line for 0.2% and model class for 2.8%. These shares are not directly comparable: a 79-level factor absorbs 29.7% of the variance of relabelled data where a three-level factor absorbs 0.8%. Against a null that permutes protein labels between datasets while preserving dataset-by-model blocks, the excess attributable to protein identity was +7.2 points ($p = 0.003$). A direct test is more informative than either share: the same protein’s log multiplier agreed between cell lines at $r = 0.586$ (Spearman 0.594, $p = 0.0001$) over 40 protein-by-model pairs. Cell line was not detectable, although with fifteen proteins measured in both lines the design has little power. Model class was small but detectable ($p = 0.023$), consistent with SpliceBERT’s lower multiplier, and robust to the censoring induced by requiring both arms positive.

We do not decompose the span into a compression term and a residual protocol effect. That decomposition requires transplanting a model’s discriminability increment across baselines, and its central assumption, that the increment is invariant to the baseline against which it is measured, is violated in these data. Across a grid of twelve specifications, 4 of 6 sign tests held for the 4-mer, 6 of 6 for the convolutional network and 1 of 6 for SpliceBERT; the log-odds link reverses the sign entirely, which is the behaviour Mood [12] describes for odds ratios under unobserved heterogeneity. With three arms the residual’s sign follows the direction of transplant, +0.0452 carrying an increment from a high-baseline arm to a low-baseline one and -0.0259 in reverse. We report the raw contrast, which requires no transplant, link or transportability assumption.

Two provenance limitations apply to this subsection. For 20 of the 94 datasets in the dinucleotide arm the retained neural scores were produced under a partition that is not chromosome-grouped. Fold sizes were preserved, so count-based checks do not detect it. Up to 23 chromosomes appear in a single fold, and up to 44.5% of rows have a same-strand neighbour within 1 kb assigned to a different fold, against zero under the study folds and zero in the GC and bias-aware arms. The 4-mer is refit on the study folds for every dataset and is unaffected. Restricted to the 74 correctly partitioned datasets the contrasts were +0.0436, +0.0506 and +0.0878, all within the protein-clustered interval half-widths of the full-panel values, which we report as primary for the neural models. Separately, the 4-mer entered the nested fit as a log odds and the neural models as probabilities; logistic regression is not invariant to a nonlinear transform of a covariate. Refitting the neural arms on the logit scale increased their contributions by +0.0008 (CNN) and +0.0030 (SpliceBERT) on the GC

arm and moved the contrasts by at most 0.0004, so the reported values are conservative. AUROC itself is invariant to monotone transforms.

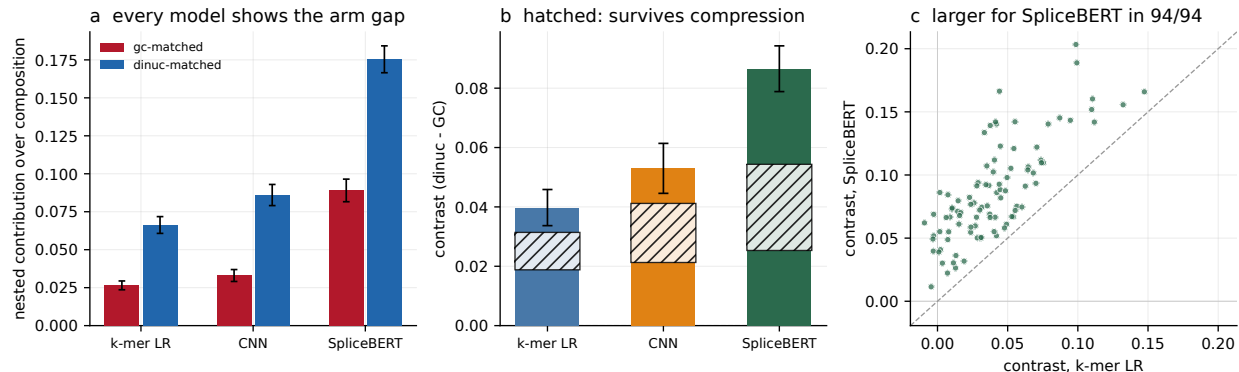


Figure 4: Nested contribution in each composition-matched arm for a 4-mer logistic regression, a 7089-parameter convolutional network and a fully fine-tuned 19.78 M-parameter SpliceBERT, on identical rows and folds, $n = 94$ datasets. Intervals are 95% percentile intervals from a bootstrap over the 79 proteins, taking all datasets of each sampled protein (4000 draws).

2.5 Replication on an independent benchmark

We applied the same decomposition to the negative sets released by Horlacher et al. [4], using their positives, peak calling, negative-set construction and fold assignments, over the 45 datasets shared with our panel (Table 5, Figure 5).

Table 5: The same decomposition applied to the released negative sets of Horlacher et al. [4], $n = 45$ datasets.

negative set	composition	apparent	contribution
negative-1 (transcript background)	0.8211	0.8606	+0.0395
negative-2 (other RBPs' sites)	0.7560	0.7726	+0.0166

The span on their benchmark was 2.381, against 2.17 for the corresponding family-matched pair in our own data, the GC-versus-bias-aware comparison. Their negative-2 is constructed like our bias-aware arm, which makes that the appropriate comparison rather than our dinucleotide-versus-GC span of 2.50.

The family structure described above reproduced arm for arm. The within-arm relation between baseline and contribution was -0.645 ($p = 1.8 \times 10^{-6}$) for their composition-matched transcript-background arm and -0.187 ($p = 0.22$) for their other-RBPs'-sites arm, against -0.545 and -0.462 for our composition-matched arms and -0.122 for ours. This also accounts for the one feature of their benchmark that is anti-monotone to ours, that their negative-2 has both the lower baseline and the lower contribution, which a cross-family residual predicts and a single-family headroom relation does not.

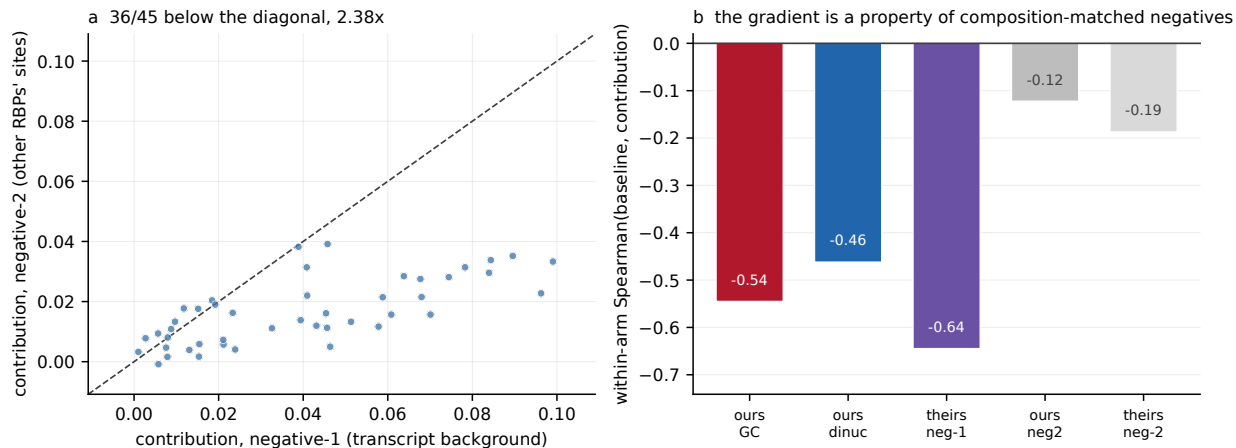


Figure 5: Replication using the released negative sets, peak calling and fold assignments of Horlacher et al. [4], over the 45 datasets shared with our panel; only the measurement is ours. (a) Nested contribution under their negative-2 against their negative-1, paired by dataset; 36 of 45 lie below the diagonal. (b) Within-arm Spearman correlation between the composition baseline and the contribution, for the composition-matched arms of both benchmarks and the other-RBPs’-sites arms of both. The relation is present in one group and absent in the other, independently in each benchmark.

2.6 Evaluation of the reporting recommendation

Reporting the composition-only AUROC alongside the headline AUROC is useful only if it improves something measurable. We tested it in the situation it is intended for, a reader holding two studies that used different protocols and wishing to compare what the models contributed (Figure 6).

On our data, normalising by headroom improved cross-protocol rank agreement in 3 of 3 protocol pairs and reduced disagreement on a common scale in 3 of 3, by 58%, 10% and 48%. Rank agreement is scale-free and cannot be flattered by normalisation. This held for all three model classes, with mean changes in rank agreement of +0.040 (4-mer), +0.094 (convolutional network) and +0.061 (SpliceBERT).

Two results qualify it. Only the GC-versus-dinucleotide improvement has an interval excluding zero (+0.051, CI +0.007 to +0.115); the remaining two pairs are consistent in direction and individually null, and the pair that reaches significance is the one in which the protocol label contributes 0.05% of variance. More importantly, the test does not replicate out of sample. On the 45 datasets of Horlacher et al. [4], rank agreement fell from 0.706 to 0.656 under normalisation and disagreement rose from 0.860 to 0.908, which are the two criteria we specified in advance as falsifying. The change in rank agreement was -0.050 (CI -0.222 to $+0.140$), so at $n = 45$ this is a failure to replicate rather than a refutation.

We therefore recommend reporting the composition-only AUROC obtained under the same protocol alongside every headline AUROC, on the grounds that it is the only summary of the protocol’s effect that a reader can act on, and we do not recommend treating any normalisation as rendering contributions comparable across protocols.



Figure 6: (a) Cross-protocol rank agreement between nested contributions before and after division by the headroom of the baseline, for each of the three protocol pairs in our data. Agreement improves in 3 of 3 pairs; only the GC-versus-dinucleotide improvement has a protein-clustered interval excluding zero. (b) The same test on the independent benchmark of Figure 5, $n = 45$: rank agreement falls and disagreement rises under normalisation, the two criteria specified in advance as falsifying. The change in rank agreement is -0.050 (-0.222 to $+0.140$), a failure to replicate rather than a refutation.

3 Discussion

How much a sequence model adds over sequence composition is not a property of the model alone. Holding the model, the positives, the folds and the estimator fixed, and varying only how the negative windows were assembled, the measured contribution of a 4-mer varied 5.4-fold, and between 3.8 and 7.6-fold across three model classes. The direction of that variation is opposite to the direction of apparent performance: the protocol that made the discrimination hardest by AUROC produced the largest contribution, and the protocol that made it easiest produced the smallest.

Two consequences for reporting follow. The composition-only AUROC obtained under the same protocol should be given alongside every headline AUROC, which costs one additional logistic regression on nineteen features. And contributions measured under different negative-set protocols should not be compared, which is a restriction on inference rather than an addition to it, and is the more consequential of the two because a substantial part of cross-study comparison in this literature is of that kind.

Apparent difficulty is not a proxy for the stringency of an evaluation. Moving to dinucleotide-matched negatives reduced apparent AUROC in every dataset examined while raising the measured contribution almost threefold, and the bias-aware protocol, which is well motivated as a means of removing compositional and accessibility artefacts, produced the easiest discrimination and the smallest measured contribution of the three. The design of that protocol is not at fault; what does not follow is the inference from a lower or higher score to how much a model has learned. Because distinct RBPs occupy compositionally distinct sites, separating one protein’s sites from another’s is largely a composition task, and a protocol that poses that task leaves a sequence model

comparatively little to add.

275 The relation between the composition baseline a protocol leaves and the contribution a model subsequently shows was strong for composition-matched negatives, both in our data and independently in that of Horlacher et al. [4], and absent for negatives drawn from the sites of other proteins. A residual difference between the two groups survived adjustment for the baseline. This suggests that what matters for a benchmark designer is less the stringency of the matching than
280 what the negative window is: contributions appear comparable within a group of protocols that draw negatives from the same kind of object, and not between such groups. This structure was, however, recovered from the 4-mer measurement and did not reproduce for either neural model, and we therefore present it as a feature of that measurement rather than of the quantity.

Protocol dependence can be reduced by fitting a transformation but not by choosing one from
285 among the standard set. Eight standard reparameterisations reduced the span without closing it, and the smallest of them, division by the baseline’s headroom, is the two-number report expressed as a single number. An exponent that closes the span does exist, but it does not transport. The exponent fitted to our three protocols leaves an independent benchmark essentially where it started; the exponent fitted to that benchmark is 2.4 times ours; and the exponent differs by a factor of
290 1.22 between model classes measured on the same data. This is the sense in which no protocol-free measure of contribution is available. The obstacle is not that no transformation works but that a transformation which works has to be refitted for each measurement, which is precisely what a comparable quantity cannot require.

The composition baseline used here stops at order two, matching what a dinucleotide-preserving
295 shuffle would hold fixed. That is defensible, but it is a choice, and it carries more weight than its brevity in the methods suggests. Refitting the baseline at order three, adding the 63 trinucleotide frequencies, removed approximately four fifths of the 4-mer’s measured contribution and 87% of the two-arm contrast, and on the bias-aware arm the median dataset retained nothing. For a bag-of- k -mers model, most of what is reported as contribution over a composition baseline is one
300 further order of composition rather than motif recognition or any positional feature. Any study reporting an increment over a composition baseline is reporting a quantity indexed by the order of that baseline, and few state which order was used.

Two results qualify this. Protocol dependence is not an artefact of where the baseline stops: at an order-three baseline the two-arm contrast remained +0.0067, +0.0309 and +0.0545 for the
305 three model classes, all with intervals excluding zero, and the span across protocols did not collapse. And the collapse is specific to models whose features are nested within the span of the baseline. Over a trinucleotide baseline the 4-mer’s contribution was positive in 65 of 94 datasets against 94 of 94 for SpliceBERT, and after accounting for the reduced headroom the raised baseline leaves, the 4-mer lost approximately 1.4 times what SpliceBERT lost. A benchmark intended to separate
310 model classes rather than to rank them by how much short-range composition they absorb could therefore raise the order of its baseline. That recommendation carries a circularity worth stating: trinucleotide counts are linear aggregates of 4-mer counts, so an order-three baseline is structurally

unfavourable to k -mer models, and separates models by how much of their signal lies outside short-range composition rather than by capacity as such.

Horlacher et al. [4] established that negative-set construction changes apparent performance across many RBP prediction methods and proposed the bias-aware alternative. The present work does not contest that and depends on it, since their released negative sets provide our only external validation. What we add is a calibration of it: the quantity that moves is not only apparent AUROC but the increment over an explicit baseline; it moves in the opposite direction to apparent difficulty; it moves by a factor of between 3.8 and 7.6 depending on the model class, which is comparable to or larger than the differences these benchmarks exist to resolve; and no reparameterisation renders it comparable across benchmarks. The same structural problem is documented in variant-effect prediction, where the assembly of the evaluation set both inflates and reorders apparent method performance [6], and the sensitivity of genomic machine-learning evaluations to the definition of the negative class has been noted more broadly [7]. We quantify it for one well-defined estimand in one well-covered assay, and separate the part that is arithmetic from the part that is not.

Limitations

The direction of the primary contrast is implied by the design, since GC matching constrains one of the composition baseline’s fifteen degrees of freedom and dinucleotide matching constrains all fifteen; only its magnitude is informative. The bias-aware arm is not subject to this argument, constraining none of the fifteen while carrying the highest baseline.

Every magnitude reported here is indexed to a mono- and dinucleotide baseline. The protocol dependence is not so indexed.

Achieved matching is not exact matching. The GC matcher’s operative acceptance bound is 0.15 rather than its nominal 0.05, holding the nominal tolerance for 94.8% of pairs, and the dinucleotide matcher reduces composition distance 2.27-fold rather than to zero, so degrees-of-freedom statements are exact for the design and approximate for the data.

All three model classes were measured under all three protocols, but the study covers two cell lines and one assay. Whether the calibration holds for other crosslinking protocols or other organisms is untested. The transform sweep, the baseline attribution and the recommendation test are reported for all three model classes; the order-three analysis is reported for the 4-mer on 30 size-stratified datasets and for the two neural models on the two composition-matched arms.

Three provenance limitations affect the neural arms. For 20 of 94 datasets in the dinucleotide arm the retained scores came from a partition that is not chromosome-grouped, and we report the 74 correctly partitioned datasets as primary with the full panel as a sensitivity; the shifts lie within the interval half-widths. Initialisation was unseeded across the 1880 fold-runs, at a measured cost of approximately 0.001 on a panel mean against a reported half-width of approximately 0.008. The convolutional and transformer arms were trained on different accelerators.

Positive sets differ slightly between arms. Window construction is identical, but a positive is dropped when its matcher finds no acceptable negative and the matchers fail on different windows,

so the two composition-matched arms’ positive sets have a Jaccard similarity with median 0.9972 and minimum 0.9237 and are identical in 10 of 94 datasets. The negatives are therefore almost, but not exactly, the only difference between arms.

No significance filter was applied to positives and no expression filter to negatives. Every called peak is a positive, so the positive set includes weak peaks that a study applying ENCODE’s standard thresholds [3] would exclude. The expression question is bounded by the control reported above rather than removed. Region labels are 6.1% to 8.7% incorrect on the negative side by our own audit, which affects the region-matching step asymmetrically.

The verification harness asserts values rather than the provenance of the code that produced them, and the fold-partition defect above is of exactly the class that this distinction permits: it was identified by inspection rather than by any of the 649 assertions. Its manuscript audit checks only values written to three or more decimal places, so quantities written to fewer were checked manually.

Confidence intervals on a single dataset’s contribution are Wald intervals on the DeLong standard error rather than bootstrap intervals, and the Firth coefficient intervals reported in the accompanying repository resample rows rather than genes, which is narrower than gene-level clustering would give.

4 Materials and Methods

4.1 Panel selection

Candidate experiments were every released ENCODE eCLIP experiment in K562 and HepG2 (139 and 105 respectively). Where a protein had several peak files we took the one with the most biological replicates, breaking ties by ENCODE’s `preferred_default` flag, and required `output_type=peaks`, `assembly=GRCh38`, `status=released` and `file_format=bed`. Every candidate was preprocessed under both composition-matching protocols; datasets retaining at least 400 out-of-fold scored pairs were eligible.

The study panel was then defined once by the dinucleotide arm, which is the stricter control: eligible datasets were sorted by pair count with a stable sort and every second one retained, giving 95 datasets over 79 proteins. Systematic sampling by rank rather than selection of the largest N was used because AUROC correlates with dataset size at $r = 0.53$ to 0.67 in these data, so a size-truncated panel would confound panel membership with the measured quantity; the retained panel spans the 0th to 99th percentile of the candidate size distribution. Of the 95, 94 also cleared the floor in the GC arm and constitute the paired panel used for every three-protocol result, giving 282 protocol-dataset cells; the model ladder uses all 95. Panel membership was written once to a manifest that every downstream stage reads. Supplementary Table S1 lists every dataset with its ENCFE file and ENCSR experiment accession.

Fifteen proteins are assayed in both cell lines and contribute two datasets each. All headline intervals therefore resample proteins rather than datasets.

4.2 Windows

Positive windows are 101 nt centred on the midpoint of each peak interval. Peak width and all significance columns are unused: no p -value, q -value or fold-change filter was applied, so the positive set comprises every called peak. Minus-strand windows are reverse-complemented and transcribed to RNA, so every sequence is the strand the protein sees. Windows containing an ambiguous base, falling outside the assembly, duplicating an existing window by start coordinate, or carrying no region annotation are dropped and counted. Analysis is restricted to chr1 to chr22 and chrX; chrY and all scaffolds and alternates are excluded.

Region class is assigned from GENCODE v45 [5] as the first of (5'UTR, 3'UTR, CDS, non-coding exon, intron) whose interval set contains the window midpoint. Region intervals were built from all transcripts with no gene-type filter and merged, with strand disregarded at this step; introns were computed as the gaps between consecutive exons. GTF 1-based inclusive coordinates are converted to 0-based half-open.

4.3 The three negative-set protocols

Each protocol produces one negative per positive. All three use a single random stream seeded at 7 per protein.

GC-matched. Candidates are windows of the same region class on the same chromosome, at least 500 nt from any peak of the target protein, drawn with probability proportional to the length of each free interval. A candidate is accepted if its GC content is within 0.05 of the positive's. The matcher relaxes: after 25 unsuccessful draws the tolerance doubles to 0.10, and if 40 draws fail the best candidate seen is accepted provided it is within three times the nominal tolerance, that is 0.15. The operative acceptance bound is therefore 0.15 rather than the nominal 0.05. Only the target protein's own peaks are excluded, so a GC-matched or dinucleotide-matched negative may be another protein's binding site. The negative inherits the positive's strand label; the consequences are quantified in the Results.

Dinucleotide-matched. Positives are grouped by region and chromosome, and for each group a candidate pool of $\max(1500, 8n)$ windows is sampled. For each positive the nearest unused candidate is assigned greedily under the L_1 distance over the sixteen dinucleotide *counts*, searched with a k -d tree over the forty nearest neighbours. Counts rather than frequencies are used because integer distances are exactly representable, and each candidate carries an additional tiebreak column that increases strictly with genomic position and is bounded below 0.5, so it can never reorder candidates whose integer distances differ but resolves exact ties by position. Without both devices the choice among tied candidates depends on the floating-point behaviour of the k -d tree implementation and a different negative is selected for about 8% of pairs between CPU architectures. Assignment is greedy rather than optimal; achieved distances are reported so the quality of the approximation is visible. Used coordinates are tracked across all groups, because one interval can be annotated as

both a non-coding exon and a 3'UTR and so appear in two region pools. No maximum distance is imposed.

Bias-aware (other RBPs' sites). Following Horlacher et al. [4], negatives for a target are the positive windows of the other panel proteins assayed in the same cell line, sampled 1:1 without replacement, excluding any donor window within 500 nt of any of the target's own positive windows. Two deviations from Horlacher et al. [4] should be noted: our donor pool is the study panel (40 to 48 donor proteins per dataset, median 45) rather than a full ENCODE survey, and exclusion is by distance from the target's 101 nt windows rather than from its full peak intervals. Sampling is performed within fold, so every pair stays inside one cross-validation fold. These negatives are transcribed, accessible to crosslinking and strand-correct by construction, and are not composition-matched in any respect.

4.4 Achieved match quality

Achieved matching was quantified over all 456,734 pairs per arm (Table 6).

Table 6: Achieved match quality between each positive and its assigned negative, all pairs. L_1 is over the sixteen dinucleotide frequencies, on a 0 to 2 scale.

arm	med. $ \Delta GC $	p90	p99	max	within 0.05	med. L_1
GC-matched	0.0297	0.0495	0.1089	0.1486	94.8%	0.500
dinucleotide-matched	0.0198	0.0693	0.1485	0.4159	85.0%	0.220
bias-aware	0.1387	0.3169	0.4555	0.6733	20.4%	0.700

The GC matcher holds its nominal tolerance for 94.8% of pairs and its observed maximum, 0.1486, sits at the 0.15 fallback cap. And “dinucleotide-matched” means a median L_1 of 0.220 rather than zero: the matcher improves composition distance 2.27-fold relative to the GC arm but does not eliminate it, so statements about degrees of freedom controlled are exact for the design and approximate for the realisation. The bias-aware arm is not composition-matched at all, with a median GC gap 4.7 times the GC arm's.

4.5 Cross-validation

Evaluation is grouped 5-fold cross-validation over whole chromosomes. The chromosome-to-fold map is frozen once for all datasets in both cell lines, so the fifteen proteins measured in both lines are evaluated on the same chromosomes and a between-line difference cannot be a partition artefact. The map was chosen by random-restart hill climbing on peak counts only, never on labels or model output, minimising the summed squared deviation from equal mass per fold with every dataset weighted equally; it was required to beat round-robin assignment, to hold the median per-dataset maximum deviation at or below 0.05, and to place no more than half of any dataset's data in one fold. Fold i is the test fold, fold $i + 1 \bmod 5$ is validation and the remaining three train, so each

fold is test once, validation once and training three times, and no additional data is spent on model selection.

Every window is scored exactly once out of fold and all AUROCs are computed on the pooled out-of-fold score vector. Pooling rather than averaging per-fold AUROCs makes the estimate invariant to fold size; the least balanced dataset places 0.42 of its data in a single fold.

Leakage between folds was audited by exact 32-mer sharing rather than by clustered sequence identity. A 101 nt window contains 70 distinct 32-mers, and two unrelated sequences share one with negligible probability, so a shared 32-mer indicates common origin.

4.6 The composition baseline and the nested contribution

The composition baseline is a 19-column design matrix: three mononucleotide frequencies, fifteen dinucleotide frequencies and the Shannon entropy of the window’s base composition. One level is dropped from each frequency family because frequencies sum to one and retaining all of them makes the design singular; dropping a level does not change the space the block spans. Every column is standardised. GC content is not a feature: it is the spanned combination C+G, and because mononucleotide counts are marginals of dinucleotide counts the block occupies the fifteen degrees of freedom of the dinucleotide simplex. The GC-matched protocol therefore constrains one linear functional of that space and the dinucleotide-matched protocol constrains all fifteen. The baseline is refit on each protocol’s own windows and is never carried between protocols.

The nested contribution of a model is

$$\text{AUROC}(\text{composition} + \text{standardised model score}) - \text{AUROC}(\text{composition alone}),$$

Both terms were fitted by L_2 -penalised logistic regression ($C = 1$, lbfgs, at most 3000 iterations), once per fold on that fold’s training rows, and evaluated as pooled out-of-fold linear predictors on identical rows and folds. The two were compared by DeLong’s paired estimator [8], in the $O(n \log n)$ midrank form of Sun and Xu [9]. The paired estimator is necessary here because the two score vectors are fitted on overlapping training data and evaluated on identical rows, so they are strongly correlated and treating them as independent understates the variance of their difference. Confidence intervals on a single dataset’s contribution are Wald intervals on the DeLong standard error.

Both arms are evaluated out of fold. An in-sample composition AUROC compared against an out-of-fold model AUROC flatters composition sufficiently to reverse the comparison on individual datasets.

4.7 Models

4-mer logistic regression. Raw 4-mer counts over the RNA sequence (98 per window, 256 columns) into L_2 -penalised logistic regression ($C = 1$). One model per fold, out-of-fold score taken as the linear predictor. The k -mer order is 4, and the contrast is positive at every order from 3 to 6.

Convolutional network. A DeepBind-style architecture [13]: convolution (4 to 16 channels, width 12), ReLU, max-pool 4, convolution (16 to 32, width 8), ReLU, global max-pool over position, dense 32 to 64, ReLU, dropout 0.5, dense 64 to 1. 7089 parameters, all trained from scratch. Global max-pooling makes the model position-invariant, which is necessary here because exploratory analysis placed the discriminative signal approximately 15 nt off centre, varying by protein.

SpliceBERT. The pretrained nucleotide language model of Chen et al. [15], `multimolecule/splicebert`, fully fine-tuned; 19.78 M parameters, all trainable. The classification head is a mean pool over real nucleotide tokens, masking classification, separator and padding tokens, then dense to 128, ReLU, dropout 0.3, dense to 1. Weights are baked into the container image at build time.

Both neural models: AdamW, weight decay 0.01, batch size 32, binary cross-entropy, single precision, at most 12 epochs with early stopping on validation AUROC at patience 4, best checkpoint restored before test scoring. Learning rate 10^{-3} for the head and 3×10^{-5} for a fine-tuned encoder. Nominal epochs overstate the training performed: the median best epoch for SpliceBERT was 3 and most runs stopped by epoch 6 or 7, whereas the convolutional network typically ran the full 12. Initial weights were drawn before the seed was set, so initialisation is unseeded across the 940 committed fold-runs; the measured cost is a per-dataset standard deviation of 0.006 to 0.010 in the nested contribution, inducing about 0.001 on a 94-dataset mean against a reported interval half-width of about 0.008.

4.8 Inference

Because fifteen of the 79 proteins contribute two datasets each and the within-protein correlation of the primary contrast is 0.92, every headline interval is a percentile interval from a bootstrap that resamples *proteins* and takes all datasets of each sampled protein, 4000 draws. Dataset-level intervals are 1.05 to 1.23 times narrower and are not reported as headline values.

Per-dataset significance counts are additionally reported at a design effect of 1.35 applied to the DeLong standard error. That factor is the product of two quantities measured on these data: 1.10 for DeLong conditioning on two score vectors as fixed when both are in fact fitted, and 1.23 for treating spatially clustered windows as independent, estimated by a 10 kb block bootstrap. It moves the count of datasets where the 4-mer significantly helps from 80 of 94 to 72 of 94.

4.9 Reproducibility

All published values are reproduced offline by a single command against committed evidence. The harness runs 614 numeric assertions against a frozen expectations file, and the number of assertions that ran is itself asserted, so a check cannot silently skip. Two assertions are stronger than regression gates: 285 published AUROCs are rebuilt from committed per-example scores to a maximum absolute difference of 2.2×10^{-16} , and the headline contrast is rebuilt from raw sequence to 1.2×10^{-6} . Per-window out-of-fold scores for all three model classes are committed, so every cell

of the model-class comparison is recomputable from the repository alone rather than being asserted
520 against a hand-entered table.

Three measures guard against platform-dependent results: dinucleotide distances are computed on integer counts, candidate ties are broken by genomic position, and the panel sort is stable. Each was added after an observed discrepancy between CPU architectures or between runs.

Analyses used NumPy [16], SciPy [17], scikit-learn [18], PyTorch [19] and Matplotlib [20].

525 Data availability

All primary data are public. Positive windows derive from ENCODE eCLIP narrowPeak files [1, 3]; every dataset used, with its ENCFE file accession and ENCSR experiment accession, is listed in Supplementary Table S1 (95 datasets, 79 proteins, 95 experiments). Gene annotation is GENCODE v45 [5]. Cell-line expression used in the transcription control comes from four
530 ENCODE polyA RNA-seq quantification files, accessions listed in the Supplementary Methods. The external validation uses the negative sets and fold assignments released by Horlacher et al. [4] at [doi:10.5281/zenodo.10600977](https://doi.org/10.5281/zenodo.10600977), downloaded and md5-verified.

Derived data supporting every number in this paper, including per-window out-of-fold model scores for all three model classes and all per-dataset result tables, are released with the code under
535 CC BY 4.0. Intermediate window tables containing genomic sequence are not redistributed; they are regenerated from the accessions above by the pipeline in one command.

Code availability

All analysis code is available at <https://github.com/Nirmalkumar31/rbp-protocol-calibration> under the MIT licence, with `results/` and `data/evidence/` under CC BY 4.0. Every published
540 value is reproducible offline from the repository with `python scripts/verify.py --local results/tables`, which runs 614 numeric assertions against a frozen expectations file and asserts the number of assertions that ran, so a check cannot silently skip.

Use of AI tools

Generative AI (Anthropic Claude) was used as a coding assistant to implement the analysis pipeline
545 and to assist in drafting text, which the author revised. The study was conceived, designed and directed by the author, who made every analytical and interpretive decision, determined which claims the evidence supports, and is solely responsible for the content of this work. All reported values are produced by code in the accompanying repository and are reproduced by its verification harness; no reference was generated by a language model.

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Competing interests

The author declares no competing interests.

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